Introduction to Artificial Intelligence

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science focused on creating systems that can perform tasks that normally require human intelligence. These tasks include understanding language, recognising images, solving problems, making predictions, and supporting decisions.

AI systems do not all work in the same way. Some systems follow rules written by developers, while others learn patterns from examples. Machine learning is an important approach in which a model learns patterns from data and uses those patterns to make predictions on new data.

2. Machine Learning

Machine Learning (ML) is a subset of AI. Instead of programming every possible rule manually, developers provide data to a learning algorithm. The algorithm uses that data to build a model. The model can then make predictions or identify patterns in information it has not seen before.

Supervised learning uses labelled examples. For instance, a model can learn to classify emails as spam or not spam from examples that already have the correct labels. **Unsupervised learning** works with data that does not have predefined labels and can discover groups or patterns. **Reinforcement learning** trains an agent through feedback in the form of rewards or penalties.

Applications and Responsible Use of AI

3. Common Applications

AI is used in many everyday applications. Recommendation systems suggest videos, music, or products based on patterns in user activity. Navigation applications estimate travel times and suggest routes using traffic data. Banks may use machine-learning models to identify unusual transactions. Healthcare teams can use AI tools to help analyse medical images, while human professionals remain responsible for clinical decisions.

4. Generative AI and Large Language Models

Generative AI refers to systems that create new content, such as text, images, audio, or code. A Large Language Model (LLM) is a type of AI model trained to process and generate language. LLMs can answer questions, summarise documents, explain concepts, and assist with writing or coding.

A prompt is the instruction or input given to a generative AI model. Prompt engineering is the practice of designing clear instructions and providing useful context so that the model is more likely to produce a relevant response. The quality of the prompt and the information provided can affect the result.

5. Limitations and Responsible Use

AI systems can make mistakes, reflect bias in their training data, or produce confident-sounding information that is incorrect. Users should verify important outputs, protect private information, and consider fairness, transparency, and safety when designing AI applications. AI is most useful when its capabilities and limitations are understood.

Quick Review

• AI is the broad field of building systems that perform intelligent tasks.
• Machine learning is an approach in which models learn patterns from data.
• Supervised learning uses labelled examples; unsupervised learning finds patterns without predefined labels.
• Generative AI creates new content, and LLMs specialise in language.
• AI outputs should be checked, especially when decisions are important.